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Concentration and health risk assessment of per- and polyfluoroalkyl substances in cosmetic and personal care products

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ABSTRACT

Per and polyfluoroalkyl substances (PFAS) are toxicologically concerning because of their potential to bioaccumulate and their persistence in the environment and the human body. We determined PFAS levels in cosmetic and personal care products and assessed their health risks. We investigated the trends in concentrations and types of PFAS contaminants in cosmetic and personal care products before and after perfluorooctane sulfonate (PFOS) and perfluorooctanoic acid (PFOA) were added to the list of persistent organic pollutants. The total PFAS concentration ranged from 1.98 to 706.75 ng g^{-1}. The hazard quotients (HQs) for PFOA, PFOS and perfluorobutanesulfonic acid (PFBS) were lower than 1, indicating no appreciable risk to consumers. Assuming the simultaneous use of all product types and the worst-case scenario for calculations, perfluoroalkyl carboxylic acids and perfluoroalkane sulfonic acids (PFSAs) also had hazard indices lower than 1. We found that adverse effects are unlikely to occur when each type of cosmetic is used separately, or even when all product types are used together. Nevertheless, the persistence and bioaccumulation characteristics of additional PFAS present in cosmetics continue to be a cause for concern. Further research is necessary to investigate the long-term impacts of using such cosmetics and the associated risks to human health.

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PFAS; PFOS; PFOA; PFBS; cosmetics; personal care products; health risk; hazard index; systemic exposure dosage; dermal absorption

Introduction

Per and polyfluoroalkyl substances (PFAS) are a group of fluorinated compounds among which perfluorooctane sulfonate (PFOS) and perfluorooctanoic acid (PFOA) are the most commonly found.[1] The high electronegativity of fluorine atoms is reflected in the structures of PFOS and PFOA, and the carbon-fluorine link in these compounds generates a powerful dipole moment.[2] This property is extremely stable at high temperatures, resulting in resistance to breakdown through the environment, bases or oxidizers.[3] They have been widely utilized in a range of applications, including textiles, clothing, carpets, paper coatings, waterproofing agents, foam and cosmetics.[4] Cosmetic and personal care products are generally used daily by consumers. PFAS are also occasionally used in water-resistant cosmetics and personal care products because of their properties.

PFAS are released into the environment by industries and consumers. These compounds can accumulate in the environment as well as in animals and humans. They are highly stable and persistent in the environment, bioaccumulate and are toxic throughout the food chain.[5] PFAS have been detected in wastewater treatment plants,[6] foodstuffs,[7] animals,[8] cosmetic and personal care products,[9] sunscreen[10] and blood.[11] The toxicology of PFAS in humans has been extensively reviewed.[12] In addition, the half-life of these compounds in humans is long. For example, the half-lives of PFOA is 3.8 years, while that of PFOS is 5.4 years.[13] The potential health risks of PFOA and PFOS were assessed using current reference doses (RfDs) for drinking water standards.[14] However, few reports have studied PFAS contamination in consumer products.

According to the Stockholm Convention, PFOS and any of its salts or related compounds have been listed in Annex B (restricting the production and use of chemicals) since 2009, while PFOA has been listed in Annex A (elimination) since 2019.[15-18] In the European Union, PFOS was added to Annex I of Regulation (EU) 2019/1021, which states that PFOS may not be placed on the market or used as a substance except for acceptable purposes at a concentration of 0.001% by weight or less. There are restrictions on how PFOA can be used, manufactured, or sold. PFOS, PFOA, perfluorononan-1-oic acid and perfluorodecanoic acid (PFDA) are currently listed in

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Annex II as prohibited substances in relation to the use of PFAS in cosmetic products.[19]

A few studies have measured the concentration of PFAS in cosmetics in Japan, Sweden, Denmark, Canada and the United States of America (USA). These studies included cosmetic and personal care products with and without PFAS in their ingredient lists. High-performance liquid chromatography coupled with tandem mass spectrometry (HPLC-MS/MS) was used to analyze the PFAS in products, such as foundations, sunscreens, lipsticks and powder.[9,20–22]

Exposure to PFAS may be followed by several adverse health effects, such as diabetes induction, increased cholesterol levels, cardiovascular damage and immune system alterations, which depend on several factors (e.g. the amount of exposure (dose), exposure pathways, duration of exposure and chemical form).[23] Our aims were: (1) to determine PFAS contamination in cosmetic and personal care products and assess their health risks. We investigated the trends in the concentration and type of PFAS contaminants and compared our results with those of other studies to determine the trend in PFAS contamination, specifically, the concentrations in cosmetic and personal care products before and after PFOS and PFOA were added to the lists of persistent organic pollutants (PFOS in 2009 and PFOA in 2019).

Methods

Standards and reagents

A total of 10 PFAS: (Perfluorobutanoic acid [PFBA] (>99.5%), Perfluoropentanoic acid [PFPA] (97%), Perfluorohexanoic acid [PFHxA] (>97%), PFHpA (99%), PFOA (>95%), Perfluorononanoic acid [PFNA] (>95%), PFDA (>98%), Perfluoroundecanoic acid, PFUnDA (>96%), Perfluorobutanesulfonic acid [PFBS] (97%), Perfluorooctanesulfonic acid [PFOS] (>98%)) used in this study were purchased from the Wako Company (Wakayama, Japan). The mobile phase for HPLC-MS was prepared using ammonium acetate (98%) purchased from Merck (Darmstadt, Germany). Methanol (used for extraction) and acetonitrile (HPLC grade) were obtained from Merck (Germany).

Cosmetic samples

All 21 cosmetic samples were domestic and international brands produced by many manufacturers purchased from stores in Thailand. All cosmetic samples were comprised four product categories: powder, lipstick, foundation and blush on. Some sample brands were labeled as waterproof, water resistant, having nonstick properties, and sweat resistant, similar to sunscreen brands. The samples in this study had different forms, ranging from the solid phase to the semi-liquid phase (cream). All products were stored in their original packaging for analysis. The details for each sample are presented in Appendix Table A1.

Pressurized liquid extraction

Cosmetic and personal care products are semi-liquid, with high protein and organic content, which hinders their analysis. Complex matrices of cosmetic and personal care products require thorough sample cleanup before analysis to eliminate interference. Pressurized liquid extraction (PLE) was performed as previously described.[10] PLE of PFAS from cosmetic samples was performed using the Dionex ASE 200 Accelerated Solvent Extraction System (ASE). Briefly, 4–5 g of each cosmetic sample was weighed and prepared in two identical sets. Subsequently, each sample was transferred into a 33 mL stainless-steel extraction cell. Cellulose filters were placed at the bottom of the extraction cell to avoid obstruction of the metal filters by the solid particles. The cell was filled with stainless steel balls and loaded into the ASE. In this study, methanol was employed as the PLE solvent for all the extracted samples. The PLE conditions were as follows: preheating for 0 min, heating for 5 min, and a 30 min static extraction time. The conditions were then, set to 100% flush volume and one extraction cycle at 80 °C with 1,000 psi. Following the extraction process using PLE, each extracted sample (60 mL) in methanol was collected. 2 mL of the extracted sample was placed in a 2 mL centrifuge tube and centrifuged at 12,000 rpm ($r = 87$ cm, 14,031 RCF) and 5 °C for 1 h to separate lipids and remove potential matrix interference. After centrifugation, the upper clear layer of the sample (0.5 mL) was transferred into a vial for HPLC-MS/MS analysis.

Instrumental analysis

The samples were analyzed using an Agilent 1200SL HPLC system (Agilent Technologies, Waldbronn, Germany). The HPLC system was interfaced with an Agilent 6400 triple quadrupole mass spectrometer (Agilent Technologies, Santa Clara, CA). The analytical column used was an Agilent Eclipse XDB-C_{18}, 4.6 × 50 mm, 1.8 µm and Plus C_{18}, 2.1 × 100 mm, 1.8 µm. Analytical ions were analyzed in multiple reaction monitoring modes. The mobile phase was (A) 10 mM ammonium acetate (CH_3COONH_4) in ultrapure water, and (B) acetonitrile (HPLC grade). The gradient conditions were as follows: the separation process started with an initial condition of 55% (A): 45% (B), ramped to 50% (A): 50% (B) at 5.0 min, then increased to 40% (A): 60% (B) at 5.5 min, held at 40% (A): 60% (B) for 4.5 min and increased to 10% (A): 90% (B) at 15 min, at a flow rate of 0.25 mL/min. The column temperature was maintained at 40 °C and the injection volume was 10 µL. PFAS were detected using the electrospray negative ionization mode (ESI) under the following operating conditions: 10 L/min gas flow with a capillary voltage of 3,500 V and gas temperature of 300 °C. The total running time was 34 min for each sample, and all the samples were stored in a refrigerator in polypropylene vials at 4 °C.

Quality assurance and quality control

Table 1 displays the detailed parameters of each PFAS, including the acronym, number of carbons, retention time, chemical abstracts service (CAS) number, precursor ion, daughter ion and correlation coefficient. The calibration curves of the PFAS solution mixes containing five

concentrations ranging from 0.1 to 10 µg L^{-1} were obtained using acetonitrile and ultrapure water (30:70, v/v). The correlation coefficients (R^2) were greater than 0.999 for all compounds. The LOD and LOQ were calculated based on the 5 g sample used in the chemical analysis.

The repeatability and recovery of the selected method were validated using cosmetic samples for confirmation. For repeatability of the selected method, three replicates of five cosmetic samples were tested by multiple analyses of spiked cosmetic samples at concentrations of 50 and 100 ng/g. The average recoveries with relative standard deviation (% RSD) for method validation could be satisfactorily used to determine PFAS concentrations in cosmetics, which are presented in Appendix Table A2. For cosmetic samples, the average recovery range was 69.14–110.04%, with a % RSD less than 21.57%. The matrix components present in the extracted samples had the potential to cause ionization suppression or enhancement of PFAS in HPLC-MS/MS, indicating that there are some losses due to matrix effects in cosmetic and personal care products. Appendix Table A2 presents the results of the recovery analysis of cosmetic and personal care products.

Health risk assessment

The Scientific Committee for Consumer Safety (SCCS) guidelines were used to assess exposure and systematically evaluate potential risks from cosmetic and personal care products. Health risks were assessed by calculating the systemic exposure dosage (SED) and hazard quotient (HQ).[24,25] The exposure frequency was quantified to assess the risk of PFOA, PFBS and PFOS by calculating the daily dosage using Eq. (1).

$$\mathrm{SED} = \frac{\mathrm{A}\left(\mathrm{g\,d^{-1}}\right) \times 1{,}000\left(\mathrm{mg\,g^{-1}}\right) \times \frac{\mathrm{C}(\%)}{100} \times \mathrm{DA_p}(\%)/100}{\mathrm{BW}} \quad (1)$$

where:

SED (mg kg^{-1}bw d^{-1}) = Systemic Exposure Dosage

A (g d^{-1}) = Estimated daily exposure to a cosmetic product in grams per day, based on the amount applied frequency of application.

C (%) = Concentration of the ingredient under investigation in the cosmetic product, which is given in ng g^{-1} and additionally converted to % content.

DA_p (%) = Dermal absorption expressed as a percentage of the test dose (assumed to be applied under real-life conditions).

BW = Standard body weight (60 kg)

The PFOA, PFBS and PFOS concentrations in cosmetic and personal care products were determined by HPLC/MS-MS. SED was calculated based on the maximum concentration (ng g^{-1}) found in this study. The values of estimated daily exposure to cosmetic product (A) were obtained following the SCCS guidelines as shown in Appendix Table A3 of daily exposure (g d^{-1}) and multiplied by 1,000 as the unit conversion from g to mg.[25] The investigation utilized a maximal dermal absorption rate of 70% for PFOA, as suggested by Franko et al.[26] Given the absence of evidence on PFBS and PFOS, this study applied this amount to represent the maximum potential dermal absorption of these substances in a worst-case scenario.

The HQ of cosmetics is the ratio between the SED of the cosmetic and the RfDs, according to Eq. (2). The RfD value refers to the amount of exposure over the course of a lifetime that does not result in hazardous responses according to physiological guideline values. The RfD is generally expressed in units of milligrams per kilogram of bodyweight per day (mg/kg/d). The RfD values for PFOS, PFBS and PFOA were 0.00002, 0.0003 and 0.00002 mg $kg^{-1}d^{-1}$, respectively.[27,28]

$$\mathrm{HQ} = \frac{\text{Systemic Exposure Dosage}}{\text{Reference Dose}} \quad (2)$$

where:

HQ = hazard quotient

The hazard index (HI) is the sum of the HQs of the PFAS in a mixture. If the HQ and/or HI values are less than 1, no adverse health effects are expected to occur at these concentrations.

Results and discussion

Concentration of PFAS in cosmetics

The details and the overall concentrations of each sample are summarized in Table 2. Concentrations lower than the LOD were assigned a value of zero for the calculation. The dominant PFAS in cosmetics were PFBA, PFPA, PFOA and

Table 1. Analytical parameters of PFAS.

Acronym	No. of carbon	RT	Cas No.	Precursor ion (*m/z*)	Daughter ion (*m/z*)	Correlation coefficient (R^2)	Limit of detection (for 5 g samples) (ng g^{-1})	Limit of quantitation (for 5 g samples) (ng g^{-1})
PFBA	4A	3.5	375-22-4	213	169	0.9993	0.96	3.60
PFPA	5A	4.8	2706-90-3	263	219	0.9993	1.20	6.00
PFHxA	6A	7.7	307-24-4	313	269	0.9997	0.72	3.60
PFHpA	7A	10.5	375-85-9	363	319	1.0000	0.72	3.60
PFOA	8A	14	335-67-1	413	369	0.9998	1.20	3.60
PFNA	9A	17	375-95-1	463	419	0.9996	1.20	6.00
PFDA	10A	21	335-76-2	513	469	0.9994	1.20	4.80
PFUnDA	11A	23.5	2058-94-8	563	519	1.0000	0.96	2.40
PFBS	4S	8.2	375-73-5	299	80	0.9999	0.36	2.40
PFOS	8S	22	1763-23-1	499	80	0.9999	1.20	6.00

Table 2. Measured PFAS concentrations by sample groups.

Sample (*N*)	Country of origin	Compound (carbon atom), concentration ng g^{-1}												
		PFBA (4A)	PFPA (5A)	PFHxA (6A)	PFHpA (7A)	PFOA (8A)	PFNA (9A)	PFDA (10A)	PFUnA (11A)	PFBS (4S)	PFOS (8S)	Total[b] (PFAS)	Upper[c]	Lower[d]
Powder														
A	Italy	2.39[a]	29.92	21.48	1.53[a]	10.14	1.60[a]	2.82[a]	n.d.	2.23[a]	1.12[a]	73.23	88.90	67.18
B	Japan	1.86[a]	1.49[a]	n.d.	n.d.	n.d.	1.29[a]	n.d.	n.d.	n.d.	n.d.	4.64	21.96	3.36
C	Japan	1.52[a]	n.d.	n.d.	n.d.	1.22[a]	n.d.	n.d.	n.d.	8.85	n.d.	11.59	23.25	10.53
D	Indonesia	2.06[a]	7.08	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	0.48[a]	6.13	15.75	25.21	14.53
E	Thailand*	1.47[a]	1.45[a]	1.03[a]	0.86[a]	1.86[a]	1.73[a]	n.d.	n.d.	698.35	n.d.	706.75	728.11	703.87
F	Thailand	2.43[a]	n.d.	n.d.	0.96[a]	n.d.	1.37[a]	n.d.	n.d.	9.11	n.d.	13.87	28.07	11.99
Lipstick														
G	Germany	17.92	n.d	n.d	1.03	1.37[a]	2.24[a]	n.d	n.d	0.63[a]	2.47	25.66	40.07	23.39
H	Canada	2.81[a]	1.30[a]	n.d	n.d	n.d	1.42[a]	n.d	n.d	0.37[a]	n.d	5.90	24.00	3.72
I	Taiwan	1.40[a]	4.12[a]	n.d	n.d	n.d	1.31[a]	n.d	n.d	n.d	n.d	6.83	21.96	3.36
J	Taiwan	2.70[a]	1.67[a]	1.10	n.d	2.98[a]	n.d	n.d	n.d	n.d	n.d	8.45	22.44	3.60
K	U.S.A	5.46	2.92[a]	n.d	n.d	n.d	n.d	n.d	n.d	0.45[a]	n.d	8.38	21.06	7.02
Foundation														
L	Belgium	1.34[a]	10.18	n.d	n.d	n.d	n.d	n.d	n.d	0.57[a]	n.d	12.09	23.38	11.50
M	U.S.A	1.98[a]	n.d	n.d	n.d	n.d	n.d	n.d	n.d	n.d	n.d	1.98	12.36	0.96
D	U.S.A	1.68[a]	1.38[a]	n.d	0.75[a]	2.38[a]	n.d	n.d	n.d	n.d	n.d	6.19	22.44	3.60
N	Taiwan	2.85[a]	57.57	n.d	n.d	2.42[a]	1.29[a]	n.d	n.d	n.d	1.50[a]	65.63	80.73	61.65
O	Japan	3.06[a]	1.73[a]	n.d	n.d	2.30[a]	n.d	n.d	n.d	n.d	9.24	16.33	27.60	12.12
E	Thailand*	6.24	5.13[a]	30.44	9.19	110.04	3.60[a]	n.d	9.75	n.d	12.07	186.46	191.29	180.13
Blush on														
P	Japan	1.73[a]	n.d	3.67	n.d	19.59	n.d	n.d	n.d	n.d	n.d	24.99	33.70	24.22
D	Indonesia	2.39[a]	3.13[a]	n.d	n.d	n.d	n.d	n.d	n.d	1.07[a]	n.d	6.59	19.20	2.52
C	Japan	2.46[a]	6.59	n.d	n.d	n.d	n.d	n.d	n.d	n.d	n.d	9.05	17.75	7.55
E	Thailand	1.39[a]	1.62[a]	n.d	n.d	2.18[a]	n.d	n.d	n.d	25.50	n.d	30.69	44.70	28.38

Note: Products were produced in 2012–2014;
*Thailand (made in Thailand by companies under Japanese control).
LOD: limit of detection; LOQ: limit of quantitation (the results of LOD and LOQ are shown in Table 1).
n.d. Concentration levels under the LOD are described as not detected (n.d). The LOD varied according to the 5 g samples used in the analysis.
[a]Concentration levels under the LOQ and over the LOD are presented as noted. The LOQ varied according to the 5 g samples used in the analysis.
[b]Total concentration was calculated by the sum of the PFAS concentrations; summation was based on the levels shown in the table, while n.d. values were assigned a zero value for the calculation.
[c]Upper boundary; concentrations lower than the LOQ and higher than the LOD were assigned the LOQ for calculation. Concentrations lower than the LOD were assigned the LOD for calculation.
[d]Lower boundary; concentrations lower than the LOQ and higher than the LOD were assigned the LOD for calculation. Concentrations lower than the LOD were assigned a zero for calculation.

PFBS, which were present in high concentrations in most samples. PFNA and PFOS were detected in some of the powder, lipstick and foundation samples. There was a large variation in the concentration of PFAS among all cosmetic samples. The highest detected concentrations of PFAS were as follows: PFBS, 698.35 ng g^{-1}; PFOA, 110.04 ng g^{-1}; and PFPA, 57.57 ng g^{-1}. The highest PFAS concentration was found in brand E (powder, foundation, and blush on), which contained the highest PFBS concentration but had quite low levels of PFOS compared to those of other PFAS. These results suggest that the detection of short-chain PFAS indicates a substitution process within industrial PFAS applications to avoid PFOS use.[29] This can be attributed to the production date, as all cosmetic samples were manufactured between 2012 and 2014 after the Stockholm Convention's restriction of PFOS production and utilization in 2009.[30] It should be noted that the highest PFAS concentration was detected in brand E for all categories (powder, foundation and blush on), except for the lipstick from brand E, for which samples were not available. However, several PFAS were detected at high levels in all brands. Brands A (powder), G (lipstick), N (foundation) and P (blush on) contained higher concentrations than other brands. PFPA was the most common chemical found in cosmetic samples, brand N (foundation) and brand A (powder) at 57.57 ng g^{-1} and 29.92 ng g^{-1}, respectively. Brand G (lipstick) had the highest PFBA concentration (17.92 ng g^{-1}). PFOA was the main contaminant detected in brand P (blush on) with a concentration of 19.59 ng g^{-1}. PFAS composition significantly changed over time, with the replacement of long-chain PFAS with short-chained PFAS and a reduction in PFOS levels through replacement with other compounds, such as PFBS.[31] In addition, when comparing PFAS levels in samples of the same brand, cosmetic samples from brand D (foundation, powder and blush on) were produced in different countries (USA and Indonesia) but the concentrations did not show much difference. It is possible that PFAS concentrations depend on the regulatory levels set by the producer.

The PFAS concentration of cosmetic samples from Thailand (under control by the parent company outside Thailand) was higher than that of samples from other countries, and the range of PFAS concentrations in samples from other countries did not differ significantly. Regulations for contamination in developing countries are unavailable. Nevertheless, we found that brand G (lipstick) samples, which were made outside Thailand, still contained higher total PFAS concentrations than the other lipstick samples, specifically PFBA, PFHpA, PFNA, PFBS and PFOS. The highest concentration of each PFAS was found mostly in Asian cosmetic samples, which may be due to differences in the quantitative use of each PFAS and the regulations to control the use of PFAS in each country.[32] According to Fujii et al.[9] another reason for the

difference in PFAS concentrations between cosmetics produced in Asia and other continents may be due to differences in waterproof ingredient recipes as well as the correlation with the biotransformation or degradation of PFAS that occurs through hydrolysis and oxidation, which may result in a small quantity of PFAS remaining in the cosmetic sample.

High levels of PFAS were detected in almost all powder samples, with the greatest concentrations of PFBS observed in the powder from brand E. Lipstick samples had the highest PFBA concentration, whereas other PFAS were detected at low levels when compared to samples from other categories. The dominant PFAS in the foundation samples was PFOA, which was at a high level compared to that of the other PFAS. In the blush on samples, PFBS and PFOA were present at high concentrations.

Compared to other PFAS, PFBS, PFOA, PFPA and PFBA were detected at high concentrations in cosmetics. PFOS was generally found at low levels, which we believe is due to changes in manufacturing, because all uses of PFOS were prohibited. In contrast, PFBS was found at high levels because long-chain (C8) fluorinated compounds were substituted with short-chain compounds.[33] Since the United States Environmental Protection Agency (USEPA), together with the fluorine industry, launched a program aimed at eliminating PFOA and its related chemicals from the US market no later than 2015 (the 2010/2015 PFOA stewardship program), PFBA has been used to replace PFOA.[34] However, we found that PFOA levels were still high in the samples examined. Nevertheless, the use of PFOA and related compounds has been decreasing since 2015, and the market volume is probably lower than in the past, including direct and indirect sources.

Brand E (powder and blush on) products were labeled as waterproof, which is consistent with the study by Kissa, who showed that PFAS can be used as a water-oil repellent.[35] Although brand F (powder) products were also labeled as waterproof, their PFAS levels were lower than those of brand A (powder) products, which were not labeled as waterproof. The PFAS concentration in lipstick from brand K, which was labeled as waterproof, was lower than that in the samples from brand G, although these were not labeled as waterproof. The lack of waterproofness indicators in the labels did not therefore mean that the products contained low PFAS levels, but that the cosmetic samples labeled as waterproof did have high PFAS levels, as shown in Figure 1.

All samples, irrespective of whether they were sold as waterproof, had detectable levels of PFAS. This is because PFAS have been used in a wide range of consumer applications and products,[29] for example, as surfactants, emulsifiers, wetting agents and coatings.[36] This suggests that PFAS are present in cosmetic products.

Concentration of PFAS in different cosmetic categories

The concentrations of all substances in the different categories of cosmetic samples, including powder, lipstick, foundation, and blush on, are shown in Figure 2. PFBS was the major PFAS in the powder category, for which the highest concentration found was 119.83 ng g^{-1}. Other PFAS were found at low levels in the powder category, especially PFHpA, PFDA and PFUnA, which were found at less than 1 ng g^{-1}. PFOS is used in consumer products as a surface active agent to repel both water and oil.[35] This property is one of the requirements of the powder production process. However, PFOS has been phased out; therefore, PFBS has been introduced as a replacement for PFOS.[37] That is why PFBS was the most common PFAS detected in powder samples and the one at the highest concentration.

For the lipstick samples, PFBA, PFPA and PFNA were the top three compounds in terms of concentration. Overall, PFAS levels were lower in lipstick samples than in the other categories, except for PFBA. PFBA was predominant in lipsticks because the majority of lipstick compositions used wax.[38] Since the USEPA aimed to eliminate PFOA and its related chemicals from the US market in 2015, PFBA may have been used to replace PFOA because of its similar properties. PFOA was the PFAS found at the highest concentrations in the foundation products, and the highest concentration found was 110.04 ng g^{-1}. According to a previous report PFOA is primarily used as an emulsifier. Emulsifiers are contained in the foundations for several industrial uses.[39,40] Hence, the foundation samples had high PFCAs levels. In the blush on category, PFBS and PFOA were the most predominant compounds, while the concentrations of PFDA, PFUnA and PFOS were slightly lower those that of other PFAS.

Trends of PFAS contamination levels in cosmetic and personal care products

Table 3 compares the results of this study with those of previous and recent studies that examined the effects of PFAS in cosmetics.[9,20,22] A previous study measured PFAS concentrations in a variety of cosmetics and discovered a high concentration of PFAS in powder and foundation samples.[9,20,22] Fujii et al.[9] discovered that powder products may contain high PFOA concentrations, ranging from undetectable (n.d.) to 1,700 ng g^{-1}, and that the PFOA concentration in foundation products ranged from n.d. to 1,500 ng g^{-1}. Schultes et al.[20] reported that the concentrations of PFOA and PFBA in foundation products were high, with ranges of n.d. –2,330 ng g^{-1} and n.d. –609 ng g^{-1}, respectively. However, PFOA and PFOS were not detected in cosmetics from the USA, whereas PFOS was detected in foundation and mascara samples from Canada.[22] Apart from the studies in Table 3, the analysis by Harris et al.[41] still identified the levels of PFOS and PFOA in cosmetic products purchased in the Canadian market in 2020. The PFOS and PFOA concentration ranges were 0–16.5 and 13.6–28,600 ng g^{-1}, respectively, with a high concentration identified in the foundation and cream for PFOS and bronzer for PFOA.

In addition, we found that the number of PFAS in the powder products decreased. The results of past and these studies on the levels of PFAS in powders demonstrate that the amount of PFAS has changed dramatically. Powder products had the widest range of PFAS contamination in 2011 (Table 3). A recent report revealed that the lowest concentration in the powder products was below the detectable

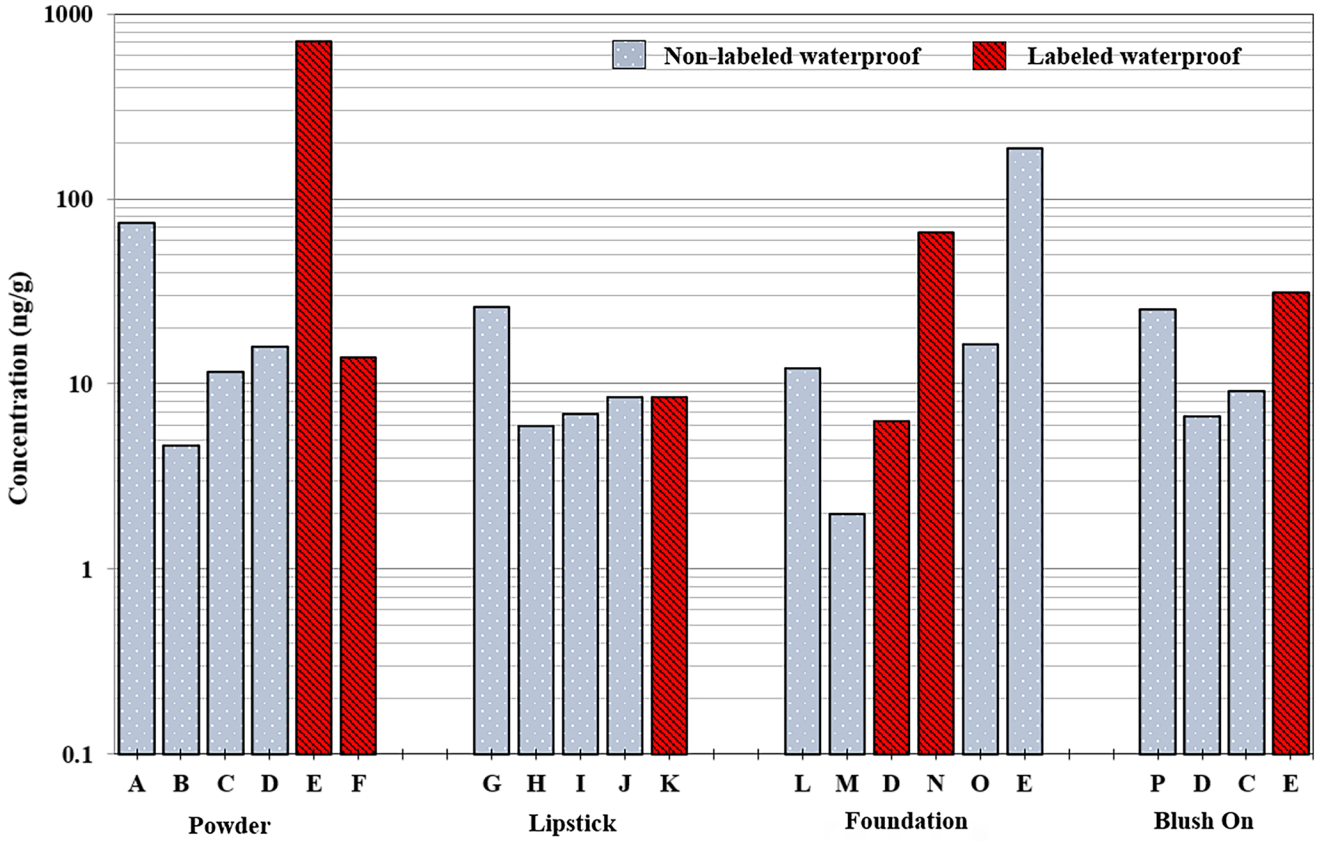


Figure 1. The median concentration of PFAS in different categories of cosmetic products.

Table 3. The concentration range of PFAS contamination in cosmetics.

Type of personal care products	Production year/ sampling year	Compound (carbon atom) of PFAS detected, sum of PFAS concentrations in range (ng g^{-1})				Country of origin	Reference
		PFBA (4A)	PFOA(8A)	PFBS (4S)	PFOS(8S)		
Powder	2011	–	n.d. –1,700	–	–	Japan, France and United States	[9]
	2012–2014	1.47–2.43	n.d.–10.14	n.d.–698.35	n.d.–6.13	Thailand, Japan, Indonesia and Italy	This study
Powders and eye shadows	2016–2017	n.d. – 84.9	n.d.–50.8	–	–	purchased PCPs from the Swedish market	[20]
Foundation	2009–2011	–	n.d.–1,500	–	–	France, Korea and Japan	[9]
	2012–2013	1.34–6.24	n.d.–110.04	n.d.–0.57	n.d.–12.07	Belgium, USA, Japan and Thailand	This study
	2016–2017	n.d. –609	n.d.–2,330	–	–	purchased PCPs from the Swedish market	[20]
	2016–2020	n.d.	–	–	–	U.S. cosmetics	[22]
	2020	–	–	–	n.d.–15.5	Canadian products	

Note: n.d. Concentration levels under the LOD are described as not detected (n.d).

level. This period of the decreasing trend was consistent with the fact that many countries were attempting to phase out the production and limit the use of long-chain PFAS, such as PFOS and PFOA, with the exception of a foundation sample from Sweden that contained 2,330 ng g^{-1} of PFOA[20] and a bronzer sample from Canada that contained 28,600 ng g^{-1} of PFOA.[41]

According to a study by Göckener et al.[42], which monitored human blood plasma samples in Germany after PFOS was added to the Stockholm Convention list and production was phased out, PFOS levels in blood plasma detected after 2010 showed a significant decreasing trend. Our investigation demonstrated that the concentration of PFOA in foundation products was higher in a previous study[9,20] than in this study. The levels of PFBA and PFOA were shown to increase in the study by Schultes et al.[20] in 2018, before PFOA was banned in 2019. Interestingly, PFOS was detected in Canadian foundation products in the study by Whitehead et al.[22] in 2021 and in the report by Harris et al.[41] even after PFOS was banned in 2009. Accordingly, Wang et al.[43] demonstrated the use of alternative fluorinated substances in various products, such as food containers, textiles, leather and fire-fighting foams. To maintain cosmetic properties, manufacturers of cosmetic and personal care products may need to change the ingredients to use fluorinated alternatives. Recent research has shown that PFAS levels have

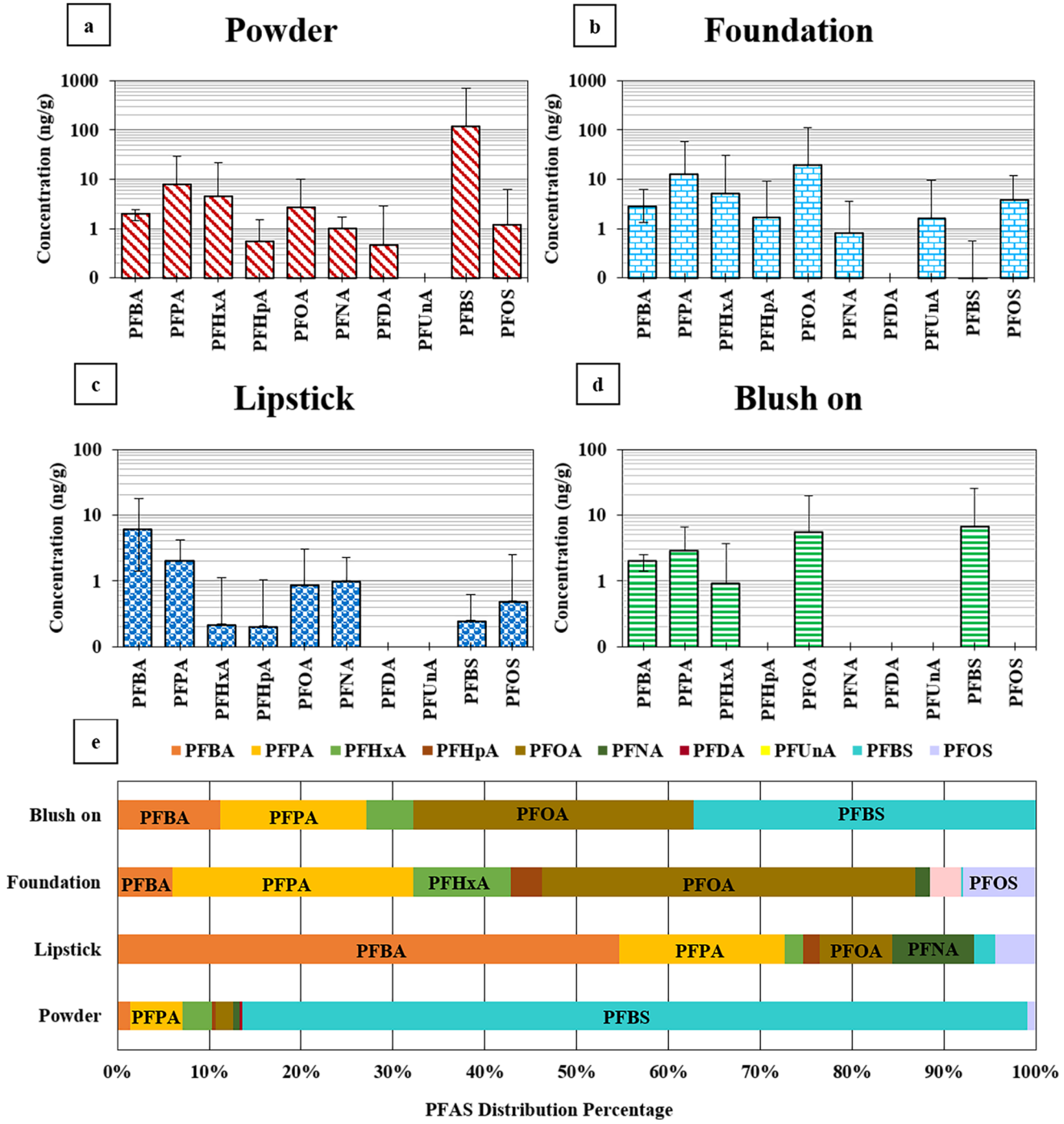


Figure 2. Concentration of each PFAS in cosmetics: a) powder, b) foundation, c) lipstick, d) blush on and e) the relative abundance of PFAS in each cosmetic.

increased. Therefore, the health risks of alternatives to PFAS must be further investigated.[20,22]

Health risk assessment

PFAS have been used in cosmetics for several years. No data are available to assess acute toxicity following exposure to high concentrations in humans. However, the EPA's Office of Water Quality has identified a health advisory for PFOS, PFOA and PFBS concentrations of 0.02, 0.004 and 2,000 ppt, respectively, for which adverse health effects from short-term exposure, as well as lifetime exposure scenarios through drinking water, are not anticipated.[44] There are no reports recommending the use of PFOS or PFOA in cosmetics. It is therefore, difficult to accurately assess human exposure to these compounds. The adverse effects of PFOS and PFOA on human health should be studied, and the risk of contamination in personal care products, including waterproof sunscreens, should be assessed.

Although commonly consumed cosmetic and personal care products are presumed to be safe,[45] the average adult uses more than one personal care product with many chemical ingredients daily. Thus, consumers are simultaneously exposed to various chemicals. There is a cumulative effect of

the daily use of multiple products. Therefore, health risk assessments of the sum of all uses should be considered.

In the above-mentioned studies, the evaluation was performed in European populations. Asian people were also considered in this assessment. Daily exposure through the application of cosmetic and personal care products for women was higher in Asia than in Europe and the USA. However, there are no current, relevant data on the frequency and amount of each type of cosmetic. These data are essential for correct exposure assessment in Asia. In this study, data on dermal absorption was still lacking. Under such circumstances, default values must be used to forecast the dermal absorption. According to a study by Franko et al.[26] the maximum dermal absorption of PFOA was 70%. Because there are still no data for PFBS and PFOS, we considered 70% as the worst-case scenario of dermal absorption for PFBS and PFOS.

The maximum concentration (ng g^{-1}) identified in this study was utilized for calculating the SED, as shown in Table 4. The highest total PFAS levels in sunscreen, powder, lipstick, foundation and blush on were selected for calculation. In almost all cases, this conservative approach results in a clear overestimation of the actual absorption, providing a high level of protection. In practice, regulatory agencies will not ask for further data if, by calculating a default value of 70% for dermal absorption, the intended use is considered acceptable.[26] The reference dose for PFOA was 0.00002 mg $kg^{-1}d^{-1}$, which was also used as the reference dose for PFCAs. The reference dose for PFOS was 0.00002 mg $kg^{-1}d^{-1}$,[27] which was used as the reference dose for perfluoroalkane sulfonic acids (PFSAs). The average adult body weight was 60 kg.[25] The result from the worst-case scenario among product categories showed that the main contributor to the high HQ value of PFCAs and PFSAs was the powder category, suggesting that powder was a major source of health risk in this study. The health risk assessment of both PFCAs and PFSAs among the different categories considering the worst-case scenario yielded an HQ value of less than 1 (Table 5), indicating that there was no risk of cancer for consumers if the use of each product was considered separately. Health risk assessment of the total amounts for all uses revealed HI values of 0.21 and 0.17 for PFSAs and PFCAs, respectively (Table 5). The evaluation of the risk associated with PFCAs and PFSAs found that the combined use of all products is likely to increase HI, but within a range that presents no hazard to human health. It should be noted that this was a preliminary study assessing dermal exposure using a limited number of samples. Further investigation is needed with a larger sample size and other potential exposure sources for PFAS. These current consumption data could be useful for safety assessors and safety agencies to protect the general population as well as at-risk subpopulations. There have been increasing concerns regarding contamination with PFAS and the effects that these may have on human health. For example, PFAS have been detected in the human breast milk.[46] In an earlier study, it was discovered that PFAS could be absorbed through the dermal layer of the human body. According to the findings of Franko et al.[26] who investigated PFOA penetration in human skin (*in vitro*), 24% of the PFOA dose that was applied had penetrated after 24 h, and the results showed that 69% of the dermal absorbable portion was present in full-thickness skin, while only 48% was present in the epidermis. Research conducted by Liu et al.[47] in the field of epidemiology revealed that blood and serum samples from pregnant women contained the highest geometric mean concentration of PFBS, which was measured at 2.767 ng mL^{-1} and had a detection rate of more than 99%. More recently, Abraham and Monien[48] reported the absorption of PFOA through the human skin using cosmetics (sunscreen), with $^{13}C_4$-PFOA identified in the plasma of volunteers at a level of 118 ng L^{-1} after 10 d and PFOA background exposure at a mean concentration of 1,400 ng L^{-1} after 115 d. According to the findings of this study, a high level of exposure to PFAS is associated with an increased risk of hypertensive

Table 4. Exposure and health risk assessments of PFOA, PFBS, and PFOS in cosmetic and personal care products using the systemic exposure dosage (SED).

	Maximum concentration (ng g^{-1}) found in this study			SED (mg kg^{-1}bw d^{-1})		
Product type	PFOA	PFBS	PFOS	PFOA	PFBS	PFOS
Sunscreen*	21.06	-	0.06	1.92E-06	-	5.47E-09
Powder	10.14	698.35	6.13	6.03E-08	4.16E-06	3.65E-08
Lipstick	2.98	0.63	2.47	2.09E-09	4.41E-10	1.73E-09
Foundation	110.04	0.57	12.07	6.55E-07	3.39E-09	7.18E-08
Blush on	19.59	25.50	-	4.57E-09	5.95E-09	-
Total	163.81	725.05	20.73	2.64E-06	4.16E-06	1.15E-07

Note: The maximum dermal absorption for PFOA is 70%.[42] Since there are no data available for PFBS and PFOS, we assumed a 70% maximum dermal absorption for PFBS and PFOS as the worst-case scenario.
*The maximum concentrations in sunscreen were obtained from the study of Keawmanee et al.[10]

Table 5. Exposure and health risk assessments of PFCAs and PFSAs in cosmetic and personal care products using the hazard index (HI).

	Maximum concentration (ng g^{-1}) found in this study		SED (mg kg^{-1}bw d^{-1})		HI	
Product type	PFCAs	PFSAs	PFCAs	PFSAs	PFCAs	PFSAs
Sunscreen*	21.06	0.06	1.92×10^{-6}	5.47×10^{-9}	9.61×10^{-2}	2.74×10^{-4}
Powder	69.88	698.35	4.16×10^{-7}	4.16×10^{-6}	2.08×10^{-2}	2.08×10^{-1}
Lipstick	21.53	3.1	1.51×10^{-8}	2.17×10^{-9}	7.54×10^{-4}	1.09×10^{-4}
Foundation	174.39	12.07	1.04×10^{-6}	7.18×10^{-8}	5.19×10^{-2}	3.59×10^{-3}
Blush on	24.99	25.5	5.83×10^{-9}	5.95×10^{-9}	2.92×10^{-4}	2.98×10^{-4}
Total	311.85	739.08	3.40×10^{-6}	4.24×10^{-6}	1.70×10^{-1}	2.12×10^{-1}

Note: Assuming the worst-case scenario for dermal absorption (70%).
*The maximum concentrations in sunscreen were obtained from the study of Keawmanee et al.[10]

disorders during pregnancy. This suggests that the PFAS in cosmetics may enter the human body.

Conclusions

PFAS have been used by several industries and consumer products because of their useful properties. We found that, among different cosmetic categories, powder products had the highest total PFAS concentration (706.75 ng g^{-1}), followed by foundation (186.46 ng g^{-1}), blush on (30.69 ng g^{-1}) and lipstick (25.66 ng g^{-1}) products. The concentrations of PFAS detected in this study were lower than those previously reported. Our results indicate that cosmetic and personal care products are the major sources of PFAS exposure in humans. Health risk assessments of cosmetics exposure of the sum of all uses must be undertaken. This study shows that the HQ value for each product type is less than 1, indicating that they do not pose an appreciable risk to consumers. When carrying out a health risk assessment of all uses (all product categories), the HI values for PFSAs and PFCAs were 0.21 and 0.17, respectively. According to this assessment, the total amount applied for all product types combined has no implications for health. However, cosmetic and personal care products in the market must be safe and provide adequate information to consumers. Further research is needed to evaluate potential human health risks.

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Ethical approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Authors' contributions

- Made substantial contributions to conception and design of the study, performed sample collection and analysis and wrote the manuscript: Keawmanee S;
- Performed data analysis and interpretation, provided content and wrote and edited the manuscript for submission: Piyaviriyakul P;
- Made a significant contribution to the methodology of sample analysis and validation of these procedures: Boontanon N;
- Reviewed and edited the manuscript for submission: Waiyarat S, Sukeesan S, Kongpran J;
- Conceptualized and designed the study; data analysis; reviewed and edited the manuscript; and provided administrative, technical, material support and funding acquisition: Boontanon SK.

Availability of data and materials

Not applicable.

Disclosure statement

All authors declared that there are no conflicts of interest.

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Appendix

Table A1. Details of cosmetic samples.

Categories	Sample (*n*)	Country of origin
Powder	4	Italy, Japan, Indonesia
	2	Thailand
Lipsticks	5	Germany, Canada, Taiwan, U.S.A
Makeup foundation	5	U.S.A, Belgium, Taiwan, Japan
	1	Thailand,
Blush on	3	Japan, Indonesia
	1	Thailand

Table A2. Recovery of PFAS from cosmetics using the optimized method.

PFAS	No. of carbon	Correlation coefficient (R^2)	% Recovery	% RSD
PFBA	4A	0.9993	90.66	10.29
PFPeA	5A	0.9993	91.42	7.17
PFHxA	6A	0.9997	95.44	13.37
PFHpA	7A	1.0000	110.04	18.40
PFOA	8A	0.9998	83.58	12.04
PFNA	9A	0.9996	84.31	15.68
PFDA	10A	0.9994	87.05	18.36
PFUnDA	11A	1.0000	70.15	19.48
PFBS	4S	0.9999	94.50	7.35
PFOS	8S	0.9999	69.14	21.57

Table A3. Estimated daily exposure levels (Safety & SCCS, 2012).

Product type	Daily exposure (g/d)
Sunscreen	7.82
Powder[1]	0.51
Lipstick	0.057
Foundation	0.51
Blush on	0.02

Note:
[1]For the powder, daily exposure data is not available. The relative daily exposure for powder was calculated by employing the value of liquid foundation daily exposure.